

A Cayley Certificate for the Mixed Gaussian S_4 Reduction

Exact level-four identities, signed-moment certification,
and a weight-seven conjecture

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A research continuation for the ProveIt project

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Abstract

We prove the mixed Gaussian identity left conjectural in the inspected ProveIt polylogarithm manuscript,

$$\sum_{n \geq 0} \frac{(-1)^n H_n}{(2n+1)^4} = \frac{58 \operatorname{Im} \operatorname{Li}_{4,1}(i, 1) + 24 \operatorname{Im} \operatorname{Li}_{3,2}(i, 1)}{7} + \frac{19\pi^5}{3584} - 2\beta(4) \log 2.$$

The proof is computer-assisted but entirely exact: a supplied rational certificate writes a specified weight-five word polynomial as a combination of 852 double-shuffle relations and one relation induced by the Cayley involution $t \mapsto (1-t)/(1+t)$. We establish the analytic validity of every relation type, including the 123 rows with one initially divergent factor. Two separately implemented verifiers check the finite word identity without floating-point arithmetic or a linear solver. The result also gives an explicit reduction of a mixed-color antisymmetric double.

For reproducible experimentation, we prove a signed-Hausdorff-moment bound for the more general coefficients $H_{rk}^{(b)}/(dk+1)^a$, with an explicit Gamma-density improvement depending on a . An exact rational Euler transform then certifies errors geometrically. It supports a new proposed weight-seven identity for S_6 with a residual enclosure of width below $3 \cdot 10^{-299}$, while rigorously excluding a misleading weight-nine near-relation. We identify the precise manuscript status changes warranted by the proof, distinguish finite relation spaces from arithmetic independence, and propose further research. The Cayley symmetry, double-shuffle framework, and Euler transform are established tools; no worldwide-priority claim is made for them or for the special-value formulas. The new contribution to the inspected project is the explicit proof certificate, its consequences, and the documented continuation.

Proof status. The S_4 identity and the analytic error bounds are proved. The S_6 identity remains a conjecture. The displayed S_8 near-relation is proved false. There is no proof-assistant formalization.

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1 Research target, provenance, and main results

1.1 The precise starting point

The project source is the collective manuscript *Polylogarithms and their Arithmetic Bridges in Analysis/Polylogarithms/docs/manuscript* of the ProveIt repository [1]. The working snapshot is commit

a2a4cf58c49c745058c40e4a6748d472a3420f18.

The audit here is targeted, not a claim to have independently verified the entire book. The relevant starting material is the mixed relation in `chapters/04-shuffle-parity.tex`, with source labels `gaussian:conj:S4` and `gaussian:eq:S4-short`. That source already proves the ordinary weight-five Gaussian shuffle reductions and explicitly leaves the mixed S_4 identity conjectural. We retain its notation:

$$H_n^{(b)} = \sum_{j=1}^n \frac{1}{j^b}, \quad H_n = H_n^{(1)}, \quad H_0^{(b)} = 0, \quad S_p = \sum_{n=0}^{\infty} \frac{(-1)^n H_n}{(2n+1)^p}, \quad (1.1)$$

$$g_{a,b} = \text{Im Li}_{a,b}(i, 1), \quad \beta(s) = \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n+1)^s}, \quad L = \log 2. \quad (1.2)$$

All indices in the special-value identities below are positive integers. The sums with $p > 1$ are absolutely convergent; the analytic arguments also cover several conditionally convergent index-one values needed by the word relations.

Theorem 1.1 (The mixed Gaussian S_4 identity). *In the strict nested-sum convention defined in Section 2,*

$$S_4 = \frac{58g_{4,1} + 24g_{3,2}}{7} + \frac{19\pi^5}{3584} - 2\beta(4)L. \quad (1.3)$$

Equivalently,

$$\text{Im Li}_{4,1}(i, -1) = \frac{51g_{4,1} + 24g_{3,2}}{7} + \frac{19\pi^5}{3584} - 2\beta(4)L. \quad (1.4)$$

The proof occupies Sections 3–5. Its finite arithmetic component is supplied in full in the accompanying `data/S4_certificate.json`; it is not an integer-relation guess.

1.2 What is added, and what is not claimed

The important extension is the use of a symmetry relation that changes intermediate depth. The single seed is the word for $\text{Li}_{4,1}(i, 1)$, but its Cayley transform contains words of depth five. Thus the calculation is not confined to an initially selected list of depth-two coordinates. Ordinary shuffle and stuffle subsequently eliminate those intermediate words. This explains the architecture of the proof without pretending that the delivered 852-row double-shuffle part is minimal.

The paper also proves the exact mixed antisymmetric identity (6.1), an all-word Cayley formula with explicit endpoint conditions, and a general signed-moment estimate. The latter produces rational enclosures for both genuine and spurious numerical relations. The S_6 formula in Conjecture 8.1 is a proposed new identity in this continuation; its very small certified residual is not a proof that it is exactly zero.

No dimension of a numerical period space is inferred. In particular, we do not prove that $g_{4,1}$ and $g_{3,2}$ are independent, that depth two is minimal, that the same basket suffices in every weight, or that the Cayley row is outside the full ideal of all standard relations. The proof uses only one explicitly defined collection of double-shuffle rows; it is not an exhaustive comparison with every possible lifted, distribution, or weight-one relation.

1.3 Relationship to established methods

Zhao's level-four work already uses the octahedral symmetry of the punctures $\{0, \infty, 1, -1, i, -i\}$ to obtain relations not detected by certain standard-relation calculations [2, 3]. The present Cayley map belongs to that established symmetry framework. Our task is the specific manuscript identity, with explicit signs, a finite certificate, and boundary-value justification. The much broader general theory of regularized relations is not being rediscovered here.

Likewise, Euler acceleration is classical [4]. The analytic contribution below is an explicit signed-measure bound for affine harmonic coefficients, including an a -dependent refinement and a simple rational implementation. The source manuscript already has signed-kernel and Chebyshev-moment evaluation methods. We do not claim that the present 2^{-N} bound improves the geometric rate of its Chebyshev method; its advantage here is that every required special value can be enclosed using rational harmonic numbers alone.

2 Analytic conventions and the level-four word algebra

2.1 Strict sums and cumulative colors

For indices $s_j \geq 1$ and colors z_j , write

$$\text{Li}_{s_1, \dots, s_d}(z_1, \dots, z_d) = \sum_{n_1 > \dots > n_d > 0} \frac{z_1^{n_1} \cdots z_d^{n_d}}{n_1^{s_1} \cdots n_d^{s_d}}. \quad (2.1)$$

At roots of unity the values used here are the ordinary convergent sums, or equivalently their radial Abel limits. It is essential that the *series colors* in (2.1) are relative colors. The corresponding nonzero letters of an iterated integral are cumulative products.

Use the forms

$$Z = \frac{dt}{t}, \quad e_j = \frac{i^j dt}{1 - ij t}, \quad j \in \mathbb{Z}/4\mathbb{Z}. \quad (2.2)$$

For a word $w = a_1 \cdots a_n$, the convention is outer-letter first:

$$I(aw; x) = \int_0^x \omega_a(t) I(w; t), \quad I(\emptyset; x) = 1, \quad I(w) = I(w; 1). \quad (2.3)$$

A nonempty word is called *admissible* if its last letter is not Z and its first letter is not e_0 . These conditions control the endpoints at zero and one, respectively. Admissible iterated integrals converge absolutely as ordered integrals, even when a corresponding outer-index-one series converges only conditionally.

If $z_j = i^{c_j}$, then

$$\text{word}((s_1, c_1), \dots, (s_d, c_d)) = Z^{s_1-1} e_{c_1} Z^{s_2-1} e_{c_1+c_2} \cdots Z^{s_d-1} e_{c_1+\dots+c_d}. \quad (2.4)$$

Expanding the geometric kernels in the disk, or differentiating recursively, proves that the integral of this word is (2.1). For example,

$$I(Z^3 e_1 e_1) = \text{Li}_{4,1}(i, 1), \quad I(Z^3 e_1 e_3) = \text{Li}_{4,1}(i, -1). \quad (2.5)$$

Using e_2 rather than e_3 as the final letter in the second expression would give a different value. This is one of the convention errors that an exact certificate must prevent.

2.2 Convergence estimates used later

For an admissible colored index word B of depth d , let B_N denote its sum with outer index at most N . Its tail obeys

$$B_N = B_\infty + O\left(\frac{(1 + \log N)^d}{N}\right). \quad (2.6)$$

For outer index at least two, comparison with $\sum_{n>N} (1 + \log n)^{d-1}/n^2$ proves the bound. For outer index one, admissibility forces its first series color c to be a nontrivial root of unity. If A_{n-1} is the inner partial sum, the coefficient $b_n = A_{n-1}/n$ satisfies

$$b_n = O((1 + \log n)^{d-1}/n), \quad b_{n+1} - b_n = O((1 + \log n)^{d-1}/n^2).$$

The second estimate follows by separating the change in $1/n$ from the new inner summand; the latter is bounded by a power of $\log n$ divided by at least n . The partial sums of c^n are bounded. Summation by parts therefore gives (2.6). This also verifies ordinary convergence and agreement with radial Abel limits.

2.3 Shuffle, stuffle, and conjugation

Let $u \sqcup v$ denote the shuffle of words, with multiplicity for repeated letters. The recursion is

$$(au) \sqcup (bv) = a(u \sqcup (bv)) + b((au) \sqcup v), \quad u \sqcup \emptyset = u. \quad (2.7)$$

Partitioning the product of two ordered simplices gives $I(u \sqcup v) = I(u)I(v)$ whenever both words are admissible.

The stuffle is formed in relative-color index coordinates. If $A = (a, c)$ and $B = (b, d)$, put $A \oplus B = (a + b, c + d \bmod 4)$ and define

$$(AU) * (BV) = A(U * (BV)) + B((AU) * V) + (A \oplus B)(U * V). \quad (2.8)$$

Sorting finite summation indices gives this identity exactly before passing to a limit. Define the raw word relation

$$\mathcal{R}(u, v) = u \sqcup v - \text{word}(\text{ind}(u) * \text{ind}(v)), \quad (2.9)$$

where the word map is extended linearly. Its integral is zero for admissible u, v . Section 3 proves the only non-admissible-factor case used in the certificate.

Complex conjugation fixes Z, e_0, e_2 and exchanges e_1, e_3 . For a word polynomial P , define $\Delta P = P - \overline{P}$. The computational encoding uses integers $-1, 0, 1, 2, 3$, where -1 means Z and $j \geq 0$ means e_j ; in particular, the integer 0 is *not* the zero form. Order encoded words lexicographically. The compressed odd projection is

$$\text{Odd}\left(\sum_w c_w w\right) = \sum_{w < \overline{w}} (c_w - c_{\overline{w}})[w]. \quad (2.10)$$

Self-conjugate words disappear. Thus

$$\text{Im } I(P) = \sum_{w < \overline{w}} (c_w - c_{\overline{w}}) \text{Im } I(w). \quad (2.11)$$

There is no hidden factor of two in (2.10); the factor appears only when P itself is already antisymmetric.

3 The one-divergence double-shuffle lemma

Lemma 3.1 (One leading $\text{Li}_1(1)$ is sufficient). *Let v be a nonempty admissible word. The formal polynomial $\mathcal{R}(e_0, v)$ has only admissible terms after like words are collected, and*

$$I(\mathcal{R}(e_0, v)) = 0. \quad (3.1)$$

No assertion about products with two divergent factors is required.

Proof. The only non-admissible word in either product expansion is e_0v . It occurs with coefficient one on each side and cancels. Indeed, a shuffle can begin with e_0 only by taking it from the first factor, since the first letter of v is not e_0 ; the remaining word must be v . In the shuffle, precisely the term obtained by putting the index-color pair $(1, 0)$ before the index word of v is divergent. Merged first indices are at least two and hence admissible.

It remains to justify that the finite parts of the two divergent terms agree. Put $B_N = \text{Li}_{\text{ind}(v)}^{(N)}$ and $B = B_\infty$. The finite cutoff of e_0v is

$$\begin{aligned} D_N &= \sum_{n=1}^N \frac{B_{n-1}}{n} = BH_N - \sum_{n=1}^N \frac{B - B_{n-1}}{n}, \\ D_N - BH_N &\longrightarrow C_B := - \sum_{n=1}^{\infty} \frac{B - B_{n-1}}{n}. \end{aligned} \quad (3.2)$$

The last series converges absolutely by (2.6). For $0 < r < 1$, the radial version is

$$I(e_0v; r) = -B \log(1 - r) - \sum_{n=1}^{\infty} \frac{r^n (B - B_{n-1})}{n}, \quad (3.3)$$

so its constant term after subtracting $-B \log(1 - r)$ is the same C_B .

The finite shuffle identity uses the product $H_N B_N$. Its difference from $H_N B$ tends to zero by (2.6). For the shuffle identity at endpoint r , the corresponding product is $-\log(1 - r)I(v; r)$. The tail bound and summation by parts imply

$$I(v; r) - B = O((1 - r)(1 + |\log(1 - r)|)^{d+1}),$$

so replacing $I(v; r)$ by B also changes the product by a quantity tending to zero. Every remaining term in the two product expansions is admissible and has its ordinary limit. Subtracting the common divergent term and then its common constant proves (3.1). $\square$

Remark 3.2. This elementary lemma is consistent with the regularization comparison in the general double-shuffle theory [3], but avoids appealing to that machinery for an unexamined divergent case. Precisely 123 of the 852 double-shuffle rows used below have first factor e_0 . The other 729 have two admissible factors. No row with two divergent factors is admitted by either verifier.

4 Cayley duality with explicit admissibility

Theorem 4.1 (The level-four Cayley involution). *Let $\phi(t) = (1 - t)/(1 + t)$ and let $\tau = -\phi^*$ act on the forms (2.2). Then*

$$\begin{array}{c|ccccc} a & Z & e_0 & e_1 & e_2 & e_3 \\ \hline \tau(a) & e_0 - e_2 & Z + e_2 & e_2 - e_3 & e_2 & e_2 - e_1 \end{array} \quad (4.1)$$

For a word $w = a_1 \cdots a_n$ define

$$\mathcal{D}(w) = \tau(a_n) \cdots \tau(a_1), \quad (4.2)$$

with multilinear expansion and ordinary concatenation on the right. Every term of $\mathcal{D}(w)$ is admissible when w is admissible, and

$$I(w) = I(\mathcal{D}(w)), \quad \mathcal{D}^2(w) = w. \quad (4.3)$$

The operator commutes with complex conjugation and preserves weight, but not necessarily depth.

Proof. The identity $\phi'(t) = -2/(1+t)^2$ gives the five pullbacks directly. For example,

$$-\phi^*Z = \frac{2dt}{1-t^2} = e_0 - e_2, \quad -\phi^*e_0 = \frac{dt}{t(1+t)} = Z + e_2.$$

For e_1 , simplifying $2i dt/((1+t)((1-i) + (1+i)t))$ gives $e_2 - e_3$; conjugation gives the e_3 row. The e_2 row is immediate.

In the ordered-simplex integral for $I(w)$, substitute $t_j = \phi(u_{n+1-j})$. This reverses the order of the variables and reverses their orientations. The resulting form at each position is $-\phi^*\omega_{a_j}$, which is exactly (4.2). Convergence permits the substitution by first restricting away from the endpoints and then taking limits.

To check that no divergent transformed word has been introduced, note that the old final letter lies in $\{e_0, e_1, e_2, e_3\}$. Every term of its transform avoids e_0 in the new first position. The old first letter lies in $\{Z, e_1, e_2, e_3\}$, and every term of its transform is a nonzero letter in the new last position. Hence all transformed words are admissible. Finally ϕ^2 is the identity, so τ^2 and $\mathcal{D}^2$ are identities. The coefficients and the map ϕ are real, which proves compatibility with conjugation. $\square$

4.1 The particular relation needed at weight five

Put

$$A = Z^3 e_1 e_1. \quad (4.4)$$

The entire additional symmetry input in the certificate is

$$I(A) = I((e_2 - e_3)^2 (e_0 - e_2)^3). \quad (4.5)$$

The powers in (4.5) are concatenation powers, not shuffle powers. The right side expands into 32 words before possible collection, all of depth five and all admissible. This is a direct specialization of Theorem 4.1, not an experimentally inferred relation.

The transformation supplies an explanation for why a calculation organized only around a small double-polylogarithm basket can miss a relation: the relation can become apparent only after passing through higher-depth words. This observation does not assert that any specific standard relation ideal is incomplete. It describes the mechanism of the certificate actually delivered here.

5 The exact finite certificate and the proof of S_4

5.1 The bridge from S_p to colored doubles

Lemma 5.1. For $p > 1$,

$$S_p = \text{Im}(\text{Li}_{p,1}(i, 1) + \text{Li}_{p,1}(i, -1)). \quad (5.1)$$

The same identity holds for $p = 1$ by convergent Abel limits.

Proof. The inner coefficient of the sum of the two doubles is $\sum_{k < m} (1 + (-1)^k)/k$. If $m = 2n + 1$, this equals H_n , because only $k = 2j$ survive. Even m have zero imaginary coefficient from i^m , whereas $\text{Im } i^{2n+1} = (-1)^n$. Absolute convergence justifies the selection for $p > 1$. The index-one case follows by first inserting a radial factor and then using convergence of the boundary values. $\square$

5.2 A target polynomial with no transcendental coefficients

Set

$$B = Z^2 e_1 Z e_1, \quad C = Z^3 e_1 e_3, \quad V = Z^3 e_1, \quad q = e_1 - e_3. \quad (5.2)$$

Here $I(B) = \text{Li}_{3,2}(i, 1)$ and $I(V) = \text{Li}_4(i)$. In the shuffle algebra define

$$T = 56\Delta C - 408\Delta A - 192\Delta B - 19q^{\sqcup 5} - 112(\Delta V \sqcup e_2). \quad (5.3)$$

Unlike the powers in (4.5), $q^{\sqcup 5}$ is the fivefold shuffle power. All coefficients in T are integers.

Lemma 5.2 (Translation of the scalar identity). *Let*

$$R_4 = 7S_4 - 58g_{4,1} - 24g_{3,2} - \frac{19\pi^5}{512} + 14\beta(4)L. \quad (5.4)$$

Then T is antisymmetric under conjugation and

$$I(T) = 16iR_4. \quad (5.5)$$

Proof. The single-value evaluations needed here are

$$I(q) = \text{Li}_1(i) - \text{Li}_1(-i) = \frac{\pi i}{2}, \quad I(e_2) = -L, \quad I(\Delta V) = 2i\beta(4).$$

They follow from $\text{Li}_1(z) = -\log(1 - z)$ on the indicated segments and the series for $\text{Li}_4(i)$. Consequently the first three terms in (5.3) contribute

$$112i \text{Im } I(C) - 816ig_{4,1} - 384ig_{3,2},$$

the fourth contributes $-19i\pi^5/32$, and the fifth contributes $224i\beta(4)L$. Substitute (5.1) with $p = 4$ and collect terms to obtain (5.5). Each term of T changes sign under conjugation; in the fourth term the shuffle power is odd. $\square$

5.3 The certificate identity

Lemma 5.3 (Exact rational word certificate). *There are 852 pairs (u_j, v_j) and rational coefficients c_j , recorded in `data/S4_certificate.json`, such that*

$$\boxed{\text{Odd}(T) = 224 \text{Odd}(A - \mathcal{D}(A)) + \sum_{j=1}^{852} c_j \text{Odd}(\mathcal{R}(u_j, v_j))}. \quad (5.6)$$

For each pair, v_j is admissible and u_j is either admissible or e_0 . All total weights are five. Exactly 541 pairs have word lengths $(1, 4)$ and 311 have lengths $(2, 3)$; exactly 123 have $u_j = e_0$.

Finite exact verification. The certificate is a list of rational numbers and finite words, not a list of approximate special values. Each double-shuffle row is regenerated by (2.7)–(2.9); the Cayley row is regenerated from (4.1). Apply (2.10) and subtract the right side of (5.6) from its left side using rational arithmetic. Every resulting coefficient is zero. The delivered primary verifier code/`verify_s4.py` performs this calculation and rejects any unallowed factor or non-admissible resulting word. Its receipt reports zero nonzero residual coordinates.

A second verifier, code/`verify_s4_independent.py`, does not import the primary algebra implementation. It enumerates shuffle placements and ordered index merges, retains full words rather than compressed odd coordinates, and checks

$$2T = 224\Delta(A - \mathcal{D}(A)) + \sum_{j=1}^{852} c_j \Delta \mathcal{R}(u_j, v_j). \quad (5.7)$$

All full-word coefficients again vanish. The two checks establish (5.6) as a finite identity over $\mathbb{Q}$. The complete coefficient data, the reconstruction rules, and executable verification kernels are part of this proof package. Their use is essential; a claimed rank or a numerical residual is not substituted for the coefficient identity. $\square$

Proof of Theorem 1.1. Each term $\mathcal{R}(u_j, v_j)$ has zero integral by ordinary double-shuffle or Lemma 3.1. The Cayley row has zero integral by Theorem 4.1. Taking imaginary parts in (5.6), using (2.11), therefore yields $\text{Im } I(T) = 0$. Lemma 5.2 gives $16R_4 = 0$. Dividing (5.4) by seven proves (1.3), and subtracting $g_{4,1}$ from (5.1) proves (1.4). $\square$

5.4 Size, provenance, and limitations of the finite computation

The sparse certificate has 853 nonzero coefficients including its one Cayley row. The largest numerator and denominator have 32 and 24 decimal digits, respectively. The target has 24 compressed odd coordinates. For orientation, there are 2000 admissible length-five words over this alphabet. Of them, 108 are fixed by conjugation, leaving 946 possible compressed odd coordinates. These counts concern words, not independent periods.

The candidate search generated weight splits (1, 4) and (2, 3) at all admissible depths, allowing the one exceptional first factor e_0 . After removal of duplicate primitive odd rows there were 1380 nonzero double-shuffle rows. A modular calculation was useful for locating a candidate; exact rational elimination then produced the certificate. Neither modular rank evidence nor the elimination software is needed to replay the proof. The verifiers consume only the fixed certificate and the elementary algebraic rules. The optional script code/`regenerate_s4.py` reconstructs the prescribed candidate rows and performs the exact elimination anew; its output is then checked by both ordinary verifiers.

No minimality statement is attached to the 852 rows, their coefficient heights, or the additional Cayley relation. In particular, this is not a proof that a suitably enlarged standard-relation system could never recover the same identity without an explicit symmetry row. A much shorter hand proof would be a valuable improvement.

6 Further proved special-value identities

Corollary 6.1 (A mixed-color antisymmetric reduction). *One has*

$$\begin{aligned} & \operatorname{Im}(\operatorname{Li}_{4,1}(i, -1) - \operatorname{Li}_{1,4}(-1, i)) \\ &= \frac{102g_{4,1} + 48g_{3,2}}{7} + \frac{79\pi^5}{10752} - 3\beta(4)L. \end{aligned} \quad (6.1)$$

Proof. The ordinary stuffle product gives

$$\operatorname{Li}_{4,1}(i, -1) + \operatorname{Li}_{1,4}(-1, i) = \operatorname{Li}_4(i) \operatorname{Li}_1(-1) - \operatorname{Li}_5(-i).$$

Its imaginary part is $-\beta(4)L + \beta(5)$. The classical odd-beta evaluation gives $\beta(5) = 5\pi^5/1536$. Subtract this sum from twice (1.4). The rational coefficient is $19/1792 - 5/1536 = 79/10752$, proving (6.1). $\square$

Corollary 6.2 (An integral evaluation). *The real logarithmic integral*

$$J_4 = \int_0^1 \frac{(-\log x)^3 \log(1+x^2)}{1+x^2} dx \quad (6.2)$$

satisfies

$$J_4 = -\frac{348g_{4,1} + 144g_{3,2}}{7} - \frac{57\pi^5}{1792} + 12\beta(4)L. \quad (6.3)$$

Proof. The generating series $\sum_{n \geq 0} H_n z^n = -\log(1-z)/(1-z)$ implies $\log(1+x^2)/(1+x^2) = -\sum_{n \geq 0} (-1)^n H_n x^{2n}$. After multiplying by $(-\log x)^3$ and integrating, absolute summability of the integrated terms gives $J_4 = -6S_4$. Now apply (1.3). $\square$

The same argument gives the integral representation used in the numerical checks:

$$S_p = -\frac{1}{(p-1)!} \int_0^1 \frac{(-\log x)^{p-1} \log(1+x^2)}{1+x^2} dx. \quad (6.4)$$

For $a, b \geq 1$, another useful representation is

$$g_{a,b} = \operatorname{Im} \frac{1}{(a-1)!} \int_0^1 \frac{(-\log x)^{a-1} i \operatorname{Li}_b(ix)}{1-ix} dx. \quad (6.5)$$

It follows by summing the inner geometric series before applying the Mellin integral for n^{-a} . For $a \geq 2$ and $b = 1$, the identity $\operatorname{Li}_{1,1}(z, 1) = \frac{1}{2} \log^2(1-z)$ gives the independent real form

$$g_{a,1} = -\frac{1}{2(a-2)!} \int_0^1 \frac{(-\log x)^{a-2}}{x} \log(1+x^2) \arctan x dx. \quad (6.6)$$

These integrals avoid a slowly convergent direct nested summation. Their floating-point quadrature is used for discovery only; the certification below uses exact rational sums instead.

7 Signed moments for affine harmonic coefficients

7.1 A general Gamma-density bound

The ordinary alternating-series remainder theorem cannot simply be applied after arbitrary transformations of harmonic coefficients. A signed-moment representation gives a different, uniform control. For a finite signed measure ν , our convention is $\|\nu\|_{\text{TV}} = |\nu|([0, 1])$, not half this quantity.

Theorem 7.1 (Affine harmonic signed moments). *Let a, r, d be positive integers and let $b > 0$ be real. Set*

$$a_k = \frac{H_{rk}^{(b)}}{(dk + 1)^a}, \quad k \geq 0, \quad (7.1)$$

and define

$$m_1 = 1, \quad m_a = \frac{(a-1)^{a-1} e^{-(a-1)}}{\Gamma(a)} \quad (a > 1). \quad (7.2)$$

There is a finite signed measure ν on $[0, 1]$, with no atom at one, such that $a_k = \int_0^1 x^k d\nu(x)$ and

$$\|\nu\|_{\text{TV}} \leq \frac{2r}{d} m_a b \zeta(b+1). \quad (7.3)$$

For $b > 1$ one also has

$$\|\nu\|_{\text{TV}} \leq 2\zeta(b). \quad (7.4)$$

In particular, for integer $b \geq 1$, usable rational bounds are

$$\|\nu\|_{\text{TV}} \leq \begin{cases} 10r/(3d), & b = 1, \\ 10/3, & b \geq 2. \end{cases} \quad (7.5)$$

Proof. Let Y have the Gamma density $y^{a-1} e^{-y}/\Gamma(a)$ and let μ be the law of $X = e^{-dY}$. Its moments are

$$\int x^k d\mu(x) = \mathbb{E}(e^{-dkY}) = (dk + 1)^{-a}. \quad (7.6)$$

In the logarithmic coordinate $s = -\log x$, the density is

$$f(s) = \frac{s^{a-1} e^{-s/d}}{d^a \Gamma(a)} \mathbf{1}_{s>0}.$$

Its maximum is m_a/d . Extended by zero to the negative half-line, its distributional derivative has total variation $2m_a/d$. For $a > 1$, this is the sum of the rise to the maximum and the fall back to zero. For $a = 1$, the jump at zero contributes $1/d$ and the decreasing exponential contributes another $1/d$.

The L^1 distance between an integrable function of bounded variation and its translate by $h \geq 0$ is at most h times the variation of its distributional derivative. This follows, for a smooth function, by integrating $f(s-h) - f(s) = -\int_{s-h}^s f'(v) dv$ and applying Fubini; approximation by smooth convolutions proves the bounded-variation case. Consequently, if $D_{\rho\mu}$ denotes pushforward under $x \mapsto \rho x$, then

$$\|\mu - D_{e^{-rt}}\mu\|_{\text{TV}} \leq \min\left(2, \frac{2m_a r}{d} t\right), \quad t > 0. \quad (7.7)$$

The elementary integral for the generalized harmonic number is

$$H_{rk}^{(b)} = \frac{1}{\Gamma(b)} \int_0^\infty \frac{t^{b-1}(1 - e^{-rkt})}{e^t - 1} dt. \quad (7.8)$$

To prove it, expand the finite geometric expression $(1 - e^{-rkt})/(e^t - 1) = \sum_{j=1}^{rk} e^{-jt}$ and integrate termwise. Define the signed measure by a total-variation-convergent integral,

$$\nu = \frac{1}{\Gamma(b)} \int_0^\infty \frac{t^{b-1}}{e^t - 1} (\mu - D_{e^{-rt}} \mu) dt. \quad (7.9)$$

Near zero, (7.7) supplies an additional factor t , so the norm of the integrand is $O(t^{b-1})$, integrable for every $b > 0$. At infinity the exponential denominator controls it. This justifies the measure integral, Fubini for bounded moment functions, and the absence of an atom at one. Combining (7.6) and (7.8) gives exactly the moments (7.1).

Using the linear bound in (7.7) yields

$$\|\nu\|_{\text{TV}} \leq \frac{2m_ar}{d\Gamma(b)} \int_0^\infty \frac{t^b}{e^t - 1} dt = \frac{2r}{d} m_a b \zeta(b+1),$$

which proves (7.3). For $b > 1$, use instead the constant bound two in (7.7), giving (7.4). For integer a , $m_a \leq 1$: with $n = a - 1 \geq 1$, this is the elementary inequality $e^n \geq n^n/n!$. Finally $\zeta(2) < 5/3$, for instance by summing n^{-2} through $n = 5$ and bounding the remaining tail by $\int_5^\infty x^{-2} dx$. Since $\zeta(b) \leq \zeta(2)$ for integer $b \geq 2$, (7.5) follows. $\square$

Remark 7.2 (Refinements and comparison with the source kernel). The full bound obtained in the proof is

$$\|\nu\|_{\text{TV}} \leq \frac{2}{\Gamma(b)} \int_0^\infty \frac{t^{b-1}}{e^t - 1} \min\left(1, \frac{m_ar}{d} t\right) dt. \quad (7.10)$$

It can improve either elementary estimate. The factor m_a in (7.3) decreases on the scale $a^{-1/2}$, by Stirling's formula. The proof therefore provides an outer-index-sensitive bound, not just a constant independent of the indices.

When $r = d$, the signed kernel already constructed in the inspected manuscript for $H_{n-1}^{(b)}/n^a$ gives another route, by pushforward under $u \mapsto u^d$. Its base positive and negative terms each have integral one and its recursion is an L^1 contraction, so it gives the additional coarse bound two. We do not treat this special-case kernel as new. Theorem 7.1 separates r and d and supplies the explicit Gamma-density refinement. The delivered interval receipts deliberately use the simpler bounds (7.5), so neither a numerical value of e nor a Gamma evaluation is trusted by the proof code.

7.2 Exact Euler acceleration for signed measures

Theorem 7.3 (Certified Euler transform). *Let $a_k = \int_0^1 x^k d\nu(x)$, where ν is a finite signed measure with $\nu(\{1\}) = 0$ and $\|\nu\|_{\text{TV}} \leq C$. Then the ordinary alternating series converges, and for every $N \geq 1$,*

$$\sum_{k=0}^\infty (-1)^k a_k = E_N + \varepsilon_N, \quad |\varepsilon_N| \leq C2^{-N}, \quad (7.11)$$

where, with the decrement difference $\delta a_k = a_k - a_{k+1}$,

$$\begin{aligned} E_N &= \sum_{j=0}^{N-1} \frac{\delta^j a_0}{2^{j+1}} \\ &= 2^{-N} \sum_{k=0}^{N-1} (-1)^k \left(2^N - \sum_{j=0}^k \binom{N}{j} \right) a_k. \end{aligned} \quad (7.12)$$

The sign convention for δ is opposite to the forward difference used in some statements of Euler's transformation.

Proof. The finite alternating kernel $\sum_{k=0}^M (-x)^k$ is bounded in absolute value by two on $[0, 1]$, and converges to $(1+x)^{-1}$ for $x < 1$. Dominated convergence with respect to $|v|$, using the absence of an atom at one, proves ordinary convergence and the formula

$$\sum_{k \geq 0} (-1)^k a_k = \int_0^1 \frac{dv(x)}{1+x}.$$

The finite geometric identity

$$\frac{1}{1+x} = \sum_{j=0}^{N-1} \frac{(1-x)^j}{2^{j+1}} + \frac{(1-x)^N}{2^N(1+x)} \quad (7.13)$$

gives (7.11), because the last kernel is bounded by 2^{-N} . Expanding $(1-x)^j$ gives the first formula for E_N . Its coefficient of a_k is $(-1)^k \sum_{j=k}^{N-1} 2^{-j-1} \binom{j}{k}$. The binomial-tail identity

$$\sum_{j=k}^{N-1} 2^{-j-1} \binom{j}{k} = 2^{-N} \sum_{j=k+1}^N \binom{N}{j}$$

follows by induction on N using Pascal's identity, and proves the second formula in (7.12). $\square$

No complete monotonicity of the harmonic sequence has been assumed. Its representing measure is signed and has total mass zero because $a_0 = 0$. The total-variation bound, rather than positivity of finite differences, is what certifies the transform.

7.3 A rational-only implementation for all constants used here

The imaginary-part selection in (2.1) gives

$$g_{a,b} = \sum_{k=0}^{\infty} \frac{(-1)^k H_{2k}^{(b)}}{(2k+1)^a}. \quad (7.14)$$

Apply Theorem 7.1 with $(r, d) = (1, 2)$ for S_p and $(r, d) = (2, 2)$ for $g_{a,b}$. The rational constants actually used are $C = 5/3$ for S_p and $C = 10/3$ for each $g_{a,b}$.

The sequences $(2k+1)^{-s}$ and $(k+1)^{-s}$ are positive moment sequences of mass one by the same Gamma integral. Thus the beta and eta values have Euler error at most 2^{-N} . Finally,

$$\pi = 4\beta(1), \quad L = \eta(1), \quad \zeta(s) = \frac{\eta(s)}{1-2^{1-s}} \quad (s > 1). \quad (7.15)$$

Every center in (7.12) and every chosen error radius is rational for the integer indices used in this paper.

The implementation maintains the binomial tail $2^N - \sum_{j=0}^k \binom{N}{j}$ incrementally. It needs $O(N)$ rational arithmetic operations for the transform once the coefficient list is available; producing the harmonic list needs $O(rN)$ rational additions. These are arithmetic-operation counts, not linear bit-complexity claims. The numerator and denominator sizes grow with N .

Exact interval arithmetic encloses products and powers without treating reused constants as independent random errors. The code multiplies interval endpoints and takes their minimum and maximum, so overestimation from dependency is harmless. To keep stored endpoints compact, it rounds outward to rationals with denominator 10^{350} , again using only integer arithmetic. Decimal strings in the receipts are display summaries; the rational lower and upper endpoints are the authoritative data.

8 A weight-seven conjecture and a rejected weight-nine relation

8.1 The surviving S_6 candidate

Conjecture 8.1 (A mixed Gaussian identity of weight seven). *With the notation (1.1)–(1.2),*

$$S_6 = -\frac{12334}{527}g_{6,1} - \frac{384}{31}g_{5,2} - \frac{2136}{527}g_{4,3} + \frac{3179\pi^7}{5713920} - 2\beta(6)L - \frac{801}{2108}\beta(4)\zeta(3). \quad (8.1)$$

The discovery search used the ordered basket

$$(S_6, g_{6,1}, g_{5,2}, g_{4,3}, \pi^7, \beta(6)L, \beta(2)\zeta(5), \beta(4)\zeta(3)). \quad (8.2)$$

At 110 working decimal digits, PSLQ with tolerance 10^{-95} , coefficient ceiling 10^{12} , and at most 10000 iterations returned

$$(-97136640, -2273402880, -1203240960, -393707520, 54043, -194273280, 0, -36910080). \quad (8.3)$$

This primitive integer vector is equivalent to (8.1); its zero coefficient on $\beta(2)\zeta(5)$ is part of the observation, not a pre-imposed omission. The decimal data are retained in the package. A separate quadrature at 180 digits supported the relation. The exact rational check below is more informative than agreement of two floating-point evaluations.

Let R_6 be the left side minus the right side of (8.1). Using $N = 1000$ in the certified Euler transform yields, after outward rounding,

$$-1.357 \cdot 10^{-299} < R_6 < 1.515 \cdot 10^{-299}. \quad (8.4)$$

The full rational interval has width less than $2.871 \cdot 10^{-299}$. This is a proved enclosure for the residual, not a proof that the residual is zero. In particular, it does not license adding (8.1) as a relation to an exact word certificate.

For orientation, the underlying values start as follows; the displayed decimals are rounded, not interval endpoints:

Quantity	Approximate value
S_6	$-0.0012883399816434530677630587571690$
$g_{6,1}$	$-0.0019411939935331449052635479564949$
$g_{5,2}$	$-0.0047581161681305610395611455068384$
$g_{4,3}$	$-0.0123729114645383875464708307852280$

All monomials in (8.1) have total weight seven. The form is therefore compatible with the weight grading. Compatibility and high-precision agreement are useful selection criteria, but neither substitutes for the missing exact reduction.

8.2 A false candidate that looked convincing at discovery precision

A broader search at weight nine used

$$\mathbf{b}_8 = (S_8, g_{8,1}, g_{7,2}, g_{6,3}, g_{5,4}, \pi^9, \beta(8)L, \beta(2)\zeta(7), \beta(4)\zeta(5), \beta(6)\zeta(3)). \quad (8.5)$$

At 110 digits, with tolerance 10^{-90} and coefficient ceiling 10^{12} , a search returned

$$\mathbf{c}_8 = (-6451250972, -4072849461, -710107844, -10984374763, -6393592639, 31875, -1955138863, 3459066102, -1034180101, -1477786143). \quad (8.6)$$

This is *not* proposed as a conjecture. Define $R_8 = \mathbf{c}_8 \cdot \mathbf{b}_8$.

Proposition 8.2 (Certified rejection of the weight-nine near-relation). *The explicit vector (8.6) is not an exact relation:*

$$-1.3274179020580459 \cdot 10^{-86} < R_8 < -1.3274179020580457 \cdot 10^{-86}. \quad (8.7)$$

In particular, $R_8 < 0$.

Proof. Apply Theorems 7.1 and 7.3 with $N = 1000$ to each harmonic sum and (7.15) to each single value. Form the indicated products and linear combination in exact interval arithmetic. The resulting rational interval, saved in `data/S8_exclusion_N1000.json`, has width below $2.080 \cdot 10^{-290}$ and is wholly contained in the interval (8.7). Both comparisons are exact comparisons of rational endpoints. Hence zero is excluded. $\square$

A separate search using 180-digit inputs, tolerance 10^{-160} , coefficient ceilings up to 10^{12} and at most 15000 iterations returned no relation in this basket. That search failure does not prove the absence of another relation, even below a height ceiling, unless the algorithm supplies and one verifies the relevant exclusion certificate. We assert only the observed failure and the rigorous rejection of the particular vector (8.6).

The numerical warning is not that PSLQ [5] is unreliable as an integer-relation algorithm. A tolerance applied to scaled coordinates can produce a vector whose *unnormalized* residual is not as small as a casual reading of the tolerance suggests. The coefficient sizes, normalizations, input errors, and recomputed residual all matter. The rational interval test checks the actual proposed identity and makes that distinction explicit.

8.3 Summary of the three residual checks

Residual	Enclosure width, upper bound	Contains zero?	Conclusion
R_4 , (5.4)	$7.109 \cdot 10^{-299}$	Yes	Consistent with the separate exact word proof.
R_6 , (8.1)	$2.871 \cdot 10^{-299}$	Yes	Strong certified numerical evidence; still conjectural.
R_8 , (8.6)	$2.080 \cdot 10^{-290}$	No	The displayed candidate is rigorously false.

The identity proof and the error certification are separate logical objects. A vanishing word polynomial proves equality for exact periods; a narrow interval containing zero does not. An interval excluding zero, however, is enough to refute a proposed equality.

9 Targeted manuscript corrections and integration

9.1 Upgrade a conjecture, not an old proof

The first integration change is a status upgrade. At the pinned snapshot, `gaussian:conj:S4` is explicitly a conjecture, supported by numerical integrals and not included in the exact shuffle certificate. That caution was appropriate. Replace its conjecture environment by a theorem and attach the Cayley proof or a precise reference to this complete proof package. The old label may be retained for compatibility even though its name contains `conj`.

The original-coordinate version in the source,

$$S_4 = \frac{4g_{4,1} - 3g_{3,2} - 9g_{2,3}}{7} + \frac{\pi^5}{224} - \frac{27\beta(2)\zeta(3)}{224} - 2\beta(4)L, \quad (9.1)$$

is the same identity. The source already proves

$$g_{2,3} = -6g_{4,1} - 3g_{3,2} - \frac{\pi^5}{1536} - \frac{3\beta(2)\zeta(3)}{32}.$$

Substitution in (9.1) gives (1.3) exactly. Thus both versions can be promoted, without re-proving or altering the existing two-shuffle elimination.

The nearby sentence saying that no functional-equation proof is supplied must be replaced, and the open-problem summary in `chapters/10-discovery.tex` must no longer list this particular S_4 identity as unresolved. Historical drafts should remain untouched as provenance; only the canonical status should change.

9.2 Qualify a non-reducibility statement

In the subsection “Family 2: antisymmetric reductions to the $(i, 1)$ algebra” in `chapters/04-depth.tex`, the prose says that at higher weight the antisymmetric parts do not reduce against the $(i, 1)$ basket alone. Read as a universal nonmembership statement, this is too strong for the search evidence described there. It is now also directly qualified by the explicit weight-five mixed antisymmetric reduction (6.1).

A suitable replacement is:

The stated searches did not recover all higher-weight antisymmetric reductions in the restricted basket. This is a limitation of those searches, not a proof of nonmembership. In particular, the Cayley certificate gives the weight-five mixed reduction displayed below.

This correction does not change the valid formal-quotient counts in other parts of the source. Those counts must continue to be described as dimensions of the specified relation systems, not as proved arithmetic dimensions.

9.3 What should remain explicitly unresolved

Do not replace S_4 by an assertion that all even-index S_{2m} have a known reduction. The status of S_6 here is conjectural, and the naive weight-nine search provided a false near-relation. The manuscript's caveats about minimal depth and numerical independence remain necessary. Similarly, no general completeness result for level-four relations is established by this certificate.

The package supplies a replacement subsection, a bibliography entry, and label-anchored editing notes under *integration/*. No changes have been pushed to the repository. The article and certificates are intended to be reviewed and integrated without overwriting historical reports or their validation receipts.

10 Further research questions

10.1 Compress the exact S_4 certificate

The most immediate problem is to replace 852 double-shuffle rows by a small human-readable chain. The single Cayley seed already identifies a geometrically meaningful source of the relation. A useful intermediate goal is a certificate grouped into a few products of lower-weight identities, with all cancellations visible. Such a compression should be measured by row count, coefficient height, and maximum intermediate depth, not merely by a short final formula.

A systematic route is to reduce the 32 depth-five words in (4.5) in a structured basis before eliminating depth-two coordinates. An exact sparse optimization or a carefully chosen Lyndon-basis normal form could produce a much smaller certificate. Any candidate compressed proof can be checked against the already verified target without recomputing period values.

10.2 Prove the S_6 identity

Conjecture 8.1 is the highest-priority new special-value target. Its coefficients are fixed, its residual is rigorously tiny, and the S_4 proof supplies a plausible mechanism. One may test Cayley images of $Z^5 e_1 e_1$ and other weight-seven Gaussian double words, together with products and lower-weight symmetry relations. The objective is an exact rational certificate, not another increment of floating-point precision.

The unexpectedly absent $\beta(2)\zeta(5)$ term suggests an elimination pattern worth understanding, but does not itself justify an all-weight formula. A structured proof should explain the denominator factors 17 and 31 that occur in the reduced coefficients, or show that they are merely basis artifacts.

10.3 Determine the correct weight-nine basket

The rejection of (8.6) leaves several logically distinct possibilities: there may be a genuine relation in the same basket with larger coefficients; additional lower-weight products may be required; or further depth-two or higher-depth coordinates may survive in the chosen relation quotient. Finite exact elimination, with an explicitly specified relation ideal, can distinguish some of these possibilities at the formal level. It cannot by itself prove transcendence or arithmetic independence of the evaluated periods.

A useful deliverable would be a complete weight-nine relation search with a reproducible basis specification, exact rank certificates, and a separate ledger of which reductions are numerical, formal, or analytic. Missing a relation at one precision should never be converted into a dimension assertion.

10.4 Compare saturated relation systems

The 1380-row search used here is deliberately limited: products of weights (1, 4) and (2, 3), with only one-divergence regularization. A fair comparison with “standard relations” in the broader literature must include the chosen distribution relations, admissible lifts of lower-weight identities, and weight-one seeds. Determine whether the specific Cayley relation lies in the saturated ideal so obtained, and record both the exact quotient and the explicit membership or nonmembership certificate in that finite model.

At higher weights, distinguish adding new geometric seed relations from multiplying already known seeds by lower-weight words. This may make symmetry-based certificates far smaller and avoid rediscovering the same structural relation in many disguises.

10.5 Improve certified arithmetic with the refined moment bounds

Theorem 7.1 gives the factor m_a and the sharper integral (7.10), but the current replay uses coarse rational constants. Implementing rational upper bounds for m_a via finite lower bounds for e^{a-1} would preserve exact arithmetic while reducing the required term count. The resulting evaluator should be compared with the manuscript’s Chebyshev and Hölder methods in bit operations, memory, and certification overhead, not just in nominal geometric rate.

Another natural extension is to products of harmonic numbers. Products of signed moment measures correspond to multiplicative convolution; this gives an immediate bound by the product of total variations, but may be far from sharp because of cancellation. Whether the Gamma-translation approach gives useful non-product bounds is a concrete analytic question.

10.6 Formalize the small trusted core

A proof-assistant implementation could first formalize the finite word identity, leaving the analytic interpretation as a separately proved layer. The data are finite lists of integers and rationals, so this stage is substantially smaller than formalizing a general multiple polylogarithm theory. A complete formalization would additionally need the iterated-integral shuffle, the convergent finite-stuffle limit, the one-divergence lemma, and the Cayley substitution with endpoint control.

The two existing Python verifiers are useful independent checks, but they are not a substitute for such kernel-level formal verification. Their agreement reduces implementation risk without changing that logical distinction.

11 Conclusion

The inspected mixed Gaussian S_4 conjecture is resolved by an explicit exact certificate whose only additional geometric input is one Cayley identity. The proof changes the status of a concrete manuscript formula, produces a mixed antisymmetric reduction, and exhibits how passing through higher-depth words can expose a relation invisible to a restricted search.

The continuation also supplies a carefully separated experimental frontier. The S_6 identity survives a rational error certificate at approximately 299 decimal places, whereas an apparently persuasive S_8 candidate is rigorously rejected. These are different mathematical outcomes and are recorded as such. The complete package preserves the finite proof data, independent verification, error bounds, and integration notes needed to extend the work without confusing numerical evidence with exact identities.

A Certificate format and the verification kernel

The JSON schema is `proveit.cayley-s4.word-certificate.v1`. A nonzero row has one of two forms:

```
{ "kind": "double_shuffle", "u": [ ... ], "v": [ ... ],
  "coefficient": "numerator/denominator" }
{ "kind": "cayley", "word": [-1,-1,-1,1,1],
  "coefficient": "224" }
```

The coefficient multiplies the *raw* row (2.9), not its primitive normalization. This convention is important because the search initially normalized duplicate rows by signed integer gcds; those normalizations have been absorbed into the published rational coefficients. The TSV file is an equivalent human-readable listing.

The essential verification loop is small:

```
residual = odd_projection(target())
for row in certificate["rows"]:
    c = Fraction(row["coefficient"])
    if row["kind"] == "double_shuffle":
        raw = double_shuffle(tuple(row["u"]), tuple(row["v"]))
    elif row["kind"] == "cayley":
        w = tuple(row["word"])
        raw = plus({w: 1}, cayley(w), -1)
    else:
        raise ValueError("Unknown relation type")
    residual = plus(residual, odd_projection(raw), -c)
if residual:
    raise ArithmeticError("Nonzero exact word residual")
```

The delivered implementation additionally validates total weight, admissibility, duplicate identifiers, and nonzero coefficients. The complete word-algebra recursions are in `code/word_algebra.py`. The alternative verifier uses placement subsets for shuffles and an explicit work list of merge paths for stuffles; it checks full antisymmetrization rather than reusing `odd_projection`.

The certificate SHA-256 digest is

```
4fd9d35fe6d1e343451034a6a35af890
07633e412a8bdafa91cb66f1dce769ed.
```

Concatenate the two displayed lines without whitespace. The digest identifies the exact JSON artifact verified here; it does not certify mathematical correctness by itself.

B Reproduction and artifact map

The exact word and interval checks require only Python 3.10 or newer and its standard library. The executed environment is recorded in `logs/environment.json`. Optional quadrature and PSLQ replay uses `mpmath`; neither it nor a linear solver is trusted by the exact verifiers. From the package root, run:

```
python code/verify_s4.py
python code/verify_s4_independent.py
python code/test_kernels.py
python code/verify_receipts.py
```

The last command recomputes all three $N = 1000$ rational enclosures and compares their endpoints exactly, without overwriting the recorded files. It also verifies the numerical inequalities printed in this paper. The optional `code/regenerate_s4.py` reconstructs the certificate by exact elimination. Complete command lines for individual enclosures, regeneration, and exploratory PSLQ searches are in `README.md`.

To rebuild the article with three serial PDFLaTeX passes, run

```
python code/build_article.py
```

The supplied PDF was compiled and rendered for review. Rebuilt PDFs can differ in metadata and need their own visual review. The manifest pins the distributed artifacts; it is not a promise of byte-identical PDF builds. In particular, a changed PDF is not covered by the original visual-review receipt.

Artifact	Purpose
article/cayley_s4.tex, PDF	Article source and reading copy.
data/S4_certificate.json, TSV	Complete rational proof certificate.
code/verify_s4*.py	Two exact word-certificate replays.
code/certified_euler.py	Rational centers, analytic radii, interval arithmetic.
data/*N1000.json	Exact rational residual enclosures.
data/*quadrature*, discovery data	Non-certified experiment records.
integration/	Replacement subsection and corrections.
logs/, MANIFEST.sha256	Executed checks, environment, and hashes.

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- [2] J. Zhao, Multiple polylogarithm values at roots of unity, *C. R. Math. Acad. Sci. Paris* **346** (2008), 1029–1032. [doi:10.1016/j.crma.2008.09.011](#). [arXiv:0810.1064](#).
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